Prelim 1
Marks 80
Time :− 3hrs mins

CBSE : X
Date :

MATHS BRIDGE ACADEMY

Lodha Signet−A,425 4th Floor, Kolshet Rd , Thane – 9892005310

SECTION A — Questions 1–20 carry 1 mark each

1. If $a = 2^2 \times 3^x$, $b = 2^2 \times 3 \times 5$, $c = 2^2 \times 3 \times 7$ and LCM(a,b,c) = 37800, then x is equal to
(A) 1 (B) 2 (C) 3 (D) 0

2. The shortest distance (in units) of the point (2, 3) from the y-axis is
(A) 2 (B) 3 (C) 5 (D) 1

3. If the lines given by $3x + 2ky = 2$ and $2x + 5y + 1 = 0$ are not parallel, then k has to be
(A) $\frac{15}{4}$ (B) $\neq \frac{15}{4}$
(C) any rational number (D) any rational number having 4 as denominator

4. A quadrilateral ABCD is drawn to circumscribe a circle. If BC = 7 cm, CD = 4 cm and AD = 3 cm, then the length of AB is
(A) 3 cm (B) 4 cm (C) 6 cm (D) 7 cm

5. If $\sec\theta + \tan\theta = x$, then $\sec\theta - \tan\theta$ will be
(A) x (B) x^2 (C) $\frac{2}{x}$ (D) $\frac{1}{x}$

6. Which one of the following is not a quadratic equation?
(A) $(x+2)^2 = 2(x+3)$ (B) $x^2 + 3x = (-1)(1-3x)^2$
(C) $x^3 - x^2 + 2x + 1 = (x+1)^3$ (D) $(x+2)(x+1) = x^2 + 2x + 3$

7. The picture shows the Olympic rings — five congruent circles of radius 1 cm each, intersecting so that the chord joining the two intersection points of two circles is also of length 1 cm. The total area of all the dotted regions (ring thickness negligible) is
(A) $4\left[\frac{\pi}{12} - \frac{\sqrt{3}}{4}\right] cm^2$ (B) $\left[\frac{\pi}{6} - \frac{\sqrt{3}}{4}\right] cm^2$
(C) $4\left[\frac{\pi}{6} - \frac{\sqrt{3}}{4}\right] cm^2$ (D) $8\left[\frac{\pi}{6} - \frac{\sqrt{3}}{4}\right] cm^2$



8. A pair of dice is tossed. The probability of not getting the sum eight is
(A) $\frac{5}{36}$ (B) $\frac{31}{36}$ (C) $\frac{5}{18}$ (D) $\frac{5}{9}$

9. If $2\sin 5x = \sqrt{3}$, $0° \leq x \leq 90°$, then x is equal to
(A) 10° (B) 12° (C) 20° (D) 50°

10. The sum of two numbers is 1215 and their HCF is 81. The number of possible pairs of such numbers is
(A) 2 (B) 3 (C) 4 (D) 5

11. If the area of the base of a right circular cone is 51 cm^2 and its volume is 85 cm^3, then the height of the cone is
(A) $\frac{5}{6}$ cm (B) $\frac{5}{3}$ cm (C) $\frac{5}{2}$ cm (D) 5 cm

12. If the zeroes of the quadratic polynomial $ax^2 + bx + c$ $(a, c \neq 0)$are equal, then
(A) c and b must have opposite signs (B) c and a must have opposite signs
(C) c and b must have same signs (D) c and a must have same signs

13. The area (in cm^2) of a sector of a circle of radius 21 cm cut off by an arc of length 22 cm is
(A) 441 (B) 321 (C) 231 (D) 221

14. If $\triangle ABC \sim \triangle DEF$, AB = 6 cm, DE = 9 cm, EF = 6 cm & FD = 12 cm, then the perimeter of $\triangle ABC$ is
(A) 28 cm (B) 28.5 cm (C) 18 cm (D) 23 cm

15. If the probability of a letter chosen at random from the letters of the word "Mathematics" being a vowel is $\frac{2}{2x+1}$, then x is equal to
(A) $\frac{4}{11}$ (B)$\frac{9}{4}$ (C) $\frac{11}{4}$ (D) $\frac{4}{9}$

16. The points A(9,0), B(9,−6), C(−9,0) and D(−9,6) are the vertices of a
(A)Square (B) Rectangle (C) Parallelogram (D) Trapezium

17. The median of a set of 9 distinct observations is 20.5. If each observation of the set is increased by 2, the median of the new set
(A) is increased by 2 (B) is decreased by 2
(C) is two times the original number (D) remains same as the original observations

18. The length of a tangent drawn to a circle of radius 9 cm from a point at a distance of 41 cm from the centre of the circle is
(A) 40 cm (B) 9 cm (C) 41 cm (D) 50 cm

DIRECTIONS (Q.19–20): A statement of Assertion (A) is followed by a statement of Reason (R). Choose the correct option:
(A) Both A and R are true and R is the correct explanation of A.
(B) Both A and R are true but R is NOT the correct explanation of A.
(C) A is true but R is false.
(D) A is false but R is true.

19. Assertion (A): The number 5^n cannot end with the digit 0, where n is a natural number.
Reason (R): A number ends with 0 if its prime factorisation contains both 2 and 5.

20. Assertion (A): If $\cos A + \cos^2 A = 1$, then $\sin^2 A + \sin^4 A = 1$
Reason (R): $\sin^2 A + \cos^2 A = 1$

SECTION B — Questions 21–25 carry 2 marks each

21. (A) The A.P. 8, 10, 12, … has 60 terms. Find the sum of the last 10 terms.

OR

(B) Find the middle term of the A.P. 6, 13, 20, …, 230.

22. If $\sin(A + B) = 1$ *and* $\cos(A - B) = \frac{\sqrt{3}}{2}, 0° < A, B < 90°$, find the measures of angles A and B.

23. If AP and DQ are medians of triangles ABC and DEF respectively, where $\triangle ABC \sim \triangle DEF$, prove that $\frac{AB}{DE} = \frac{AP}{DQ}$.

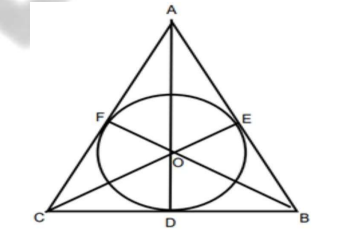
MATHS BRIDGE ACADEMY

24. (A) A horse, a cow and a goat are tied, each by ropes of length 14 m, at the corners A, B and C of grassy triangular field ABC with sides 35 m, 40 m and 50 m. Find the area of grass field that can be grazed by them.

OR

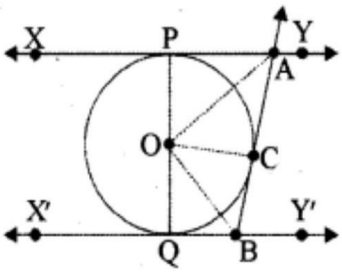
(B) Find the area of the major segment (in terms of π) of a circle of radius 5 cm, formed by a chord subtending an angle of 90° at the centre.

25. A △ABC is drawn to circumscribe a circle of radius 4 cm such that the segments BD and DC are of lengths 10 cm and 8 cm respectively. Find the lengths of the sides AB and AC, given that ar(△ABC) = 90 cm².



SECTION C — Questions 26–31 carry 3 marks each

26. In the figure, XY and X′Y′ are two parallel tangents to a circle with centre O, and another tangent AB with point of contact C intersects XY at A and X′Y′ at B. Prove that ∠AOB = 90°.



27. In a workshop, the number of teachers of English, Hindi and Science are 36, 60 and 84 respectively. Find the minimum number of rooms required, if in each room the same number of teachers (all of one subject) are to be seated.

28. Find the zeroes of the quadratic polynomial $2x^2 - \left(1 + 2\sqrt{2}\right)x + \sqrt{2}$ and verify the relationship between the zeroes and the coefficients of the polynomial.

29. (A) If sinθ + cosθ = $\sqrt{2}$ then prove that tanθ + cotθ = 2.

OR

(B) Prove that $\frac{cosA - sinA + 1}{cosA + sinA - 1} = cosecA + cotA$.

30. On a particular day, Vidhi and Unnati couldn't decide who would drive the car. They each had one coin and flipped it exactly three times, agreeing that:

1. If Vidhi gets two heads in a row, she would drive.

2. If Unnati gets a head immediately followed by a tail, she would drive.

Who has the greater probability of driving the car that day? Justify your answer.

31. (A) The monthly incomes of Aryan and Babban are in the ratio 3:4 and their monthly expenditures are in the ratio 5:7. If each saves ₹15,000 per month, find their monthly incomes.

OR

(B) Solve graphically: $2x + y = 6, 2x - y - 2 = 0$. Find the area of the triangle formed by the two lines and the x-axis.

MATHS BRIDGE ACADEMY

SECTION D — Questions 32–35 carry 5 marks each

32. A train travels a distance of 63 km at a certain average speed, then travels 72 km at an average speed 6 km/hr more than the original speed. If the total journey takes 3 hours, find the original average speed.

33. Prove that if a line is drawn parallel to one side of a triangle to intersect the other two sides in distinct points, the other two sides are divided in the same ratio.

Hence, in $\triangle PQR$, a line ℓ intersects sides PQ and PR at L and M respectively such that LM ∥ QR. If PL = 5.7 cm, PQ = 15.2 cm and MR = 5.5 cm, find the length of PM (in cm).

34. (A) From a solid right circular cone of height 6 cm and base radius 12 cm, a right circular cylindrical cavity of height 3 cm and radius 4 cm is hollowed out, the bases of cone and cylinder forming concentric circles. Find the surface area of the remaining solid, in terms of π.

OR

(B) An empty cone of radius 3 cm and height 12 cm is filled with ice-cream such that the lower $\left(\frac{1}{6}\right) th$ of the cone's volume is left empty, with a hemisphere formed on top. Find the volume of the ice-cream.

35. (A) If the mode of the following distribution is 55, find x, hence find the mean.

Class: 0-15, 15-30, 30-45, 45-60, 60-75, 75-90

Frequency: 10, 7, x, 15, 10, 12

OR

(B) A survey of heights (cm) of 51 girls of Class X gave: fewer than 140 — 4; fewer than 145 — 11; fewer than 150 — 29; fewer than 155 — 40; fewer than 160 — 46; fewer than 165 — 51. Find the median height. If the mode is 148.05, find the mean using the empirical formula.

SECTION E — Case-study questions 36–38 carry 4 marks each

36. In a class, the teacher asks every student to write an example of an A.P. Two boys, Aryan and Roshan, write the progressions −5, −2, 1, 4, … and 187, 184, 181, … respectively. Help the students answer:i. Find the sum of the common differences of the two progressions. [1]

ii. Find the 34th term of the progression written by Roshan. [1]

iii.(A) Find the sum of the first 10 terms of Aryan's progression. OR

(B) Which term of the two progressions will have the same value? [2]

37. A group of Class X students goes on a picnic. The positions of three friends — Aman, Kirti and Chahat — are given by points P(2,5), Q(4,4) and R(8,3).(i) Find the distance between P and R. [1]

(ii) Is Q the midpoint of PR? Justify by finding the midpoint of PR. [1]

(iii)(A) Find the point on the x-axis equidistant from P and Q. OR

(B) S divides PQ in the ratio 2:3. Find the coordinates of S. [2]

38. India Gate (formerly the All India War Memorial) stands near Kartavya Path, New Delhi, as a memorial to 74,187 soldiers of the Indian Army who gave their lives in the First World War. This 42 m tall structure was designed by Sir Edwin Lutyens in the style of Roman triumphal arches. A student, Shreya, of height 1 m, visited India Gate on her study tour.

i. What is the angle of elevation from Shreya's eye to top of India Gate, if she stands 41 m away from it?

ii. If Shreya observes the angle of elevation to be 60°, how far is she standing from the base of India Gate? [1]

iii.(A) If the angle of elevation changes from 45° to 30° as Shreya moves back, find the distance she moves back. OR

(B) If Shreya stands at a distance of $\frac{41}{\sqrt{3}}$m from India Gate, find the angle of elevation. [2]